

EuroSAT Land Cover Classification

Using Deep Learning

CNN from Scratch · ResNet-50 · ConvNeXT-tiny

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27,000 Satellite Images	10 Land-Cover Classes	3 Models Evaluated	96.26% Peak Test Accuracy	7,960 Min. Trainable Params
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Abstract

This project investigates satellite image classification using the EuroSAT dataset. A custom Convolutional Neural Network (CNN) was first developed as a baseline model and subsequently compared against two state-of-the-art pretrained architectures using transfer learning: ResNet-50 (Microsoft, 2015) and ConvNeXT-tiny (Facebook/Meta, 2022). The objective is to evaluate the effectiveness of transfer learning for remote sensing image classification, with focus on predictive performance, computational efficiency, and practical deployment considerations. Results demonstrate that transfer learning achieves up to 96.26% test accuracy, representing a gain of approximately 15 percentage points over the custom baseline. ConvNeXT-tiny is identified as the optimal deployment candidate, achieving 95.16% accuracy with only 7,960 trainable parameters trained over five epochs.

Keywords: satellite image classification, deep learning, convolutional neural networks, transfer learning, EuroSAT, remote sensing, ResNet-50, ConvNeXT

1. Introduction

Satellite image classification plays a critical role in modern applications such as environmental monitoring, urban planning, and agricultural analysis. However, accurately distinguishing between land cover categories remains challenging due to strong visual similarities between certain classes, complex intra-class variability, and the limited spatial resolution of publicly available datasets.

This project addresses the classification problem using the EuroSAT dataset — a widely used benchmark for remote sensing classification tasks derived from Sentinel-2 satellite imagery. The study follows a progressive modelling approach: a custom CNN baseline is first trained from scratch to establish a performance reference, and then two pretrained architectures are applied via transfer learning to assess the gains achievable by leveraging large-scale pretraining.

1.1 Objectives

The main goals of this study are:

- To establish a strong CNN baseline by training a custom architecture on EuroSAT from scratch.
- To evaluate the effectiveness of transfer learning using two architectures from different generations: ResNet-50 (2015) and ConvNeXT-tiny (2022).
- To identify the best model in terms of accuracy, computational efficiency, and practical suitability for deployment.
- To provide actionable insights on model selection for satellite image classification at scale.

2. Dataset

The EuroSAT dataset consists of 27,000 RGB satellite images of size 64×64 pixels, distributed across 10 land-use and land-cover classes derived from Sentinel-2 multispectral imagery. The dataset is publicly available and has become a standard benchmark for evaluating deep learning models in the remote sensing domain.

Dataset sources:

- EuroSAT GitHub (Helber et al.): <https://github.com/phelber/EuroSAT>
- torchvision.datasets.EuroSAT:
<https://pytorch.org/vision/stable/generated/torchvision.datasets.EuroSAT.html>

2.1 Land-Cover Classes

The ten land-cover classes covered by EuroSAT are: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. Each class is represented by between 2,000 and 3,000 images, yielding a near-perfectly balanced distribution.

2.2 Data Splits

The dataset was partitioned into training (70%, approximately 18,900 images) and validation/test (30%, approximately 8,100 images) subsets using a fixed random seed (42) to ensure full reproducibility across all experimental runs.

3. Exploratory Data Analysis

An Exploratory Data Analysis (EDA) was conducted prior to model training to characterise the dataset, identify potential modelling challenges, and inform preprocessing decisions. The analysis covered class distribution, visual inspection, pixel statistics, data quality, and inter-class relationships.

3.1 Class Distribution

Class distribution analysis confirmed that EuroSAT is well-balanced. All ten classes are approximately equally represented, each contributing roughly 10% of the total dataset. No class contains fewer than 2,000 images. This balance has two important implications: first, overall accuracy is a reliable evaluation metric without requiring class-weighted loss functions or resampling strategies; second, prediction bias towards majority classes is not a concern.

Figure 3.1 — Class distribution across EuroSAT (pie chart and count table). All classes contribute approximately 10% of the total dataset.

3.2 Image Statistics and Normalisation

Per-channel pixel statistics were computed over 2,048 training samples to establish the normalisation parameters used in all subsequent modelling stages. The results are summarised in Table 3.1.

Channel	Mean (0–255)	Std (0–255)	Mean (norm.)	Std (norm.)
Red	86.04	50.29	0.3374	0.1972
Green	95.75	34.04	0.3755	0.1335
Blue	103.00	28.79	0.4039	0.1129

Table 3.1 — Per-channel pixel statistics for the EuroSAT training split (n=2,048 samples). Values are reported in both the original 0–255 range and the normalised form applied during training.

Channel means fall in a similar range (86–103), indicating a moderately bright and well-balanced RGB representation in which no single channel dominates. Standard deviations are spread fairly evenly (29–50 per channel), confirming consistent variance across the colour space. These statistics were applied as

per-channel normalisation (zero mean, unit variance) to all three models, ensuring stable gradient flow and faster convergence during optimisation.

3.3 Visual Analysis and Inter-class Relationships

Visual inspection of individual samples and per-class mean images revealed clear differences across most classes, but also highlighted two problematic pairs:

- AnnualCrop vs. PermanentCrop: both exhibit geometric field patterns in muted greens and browns, making them visually indistinguishable at 64×64 resolution.
- Forest vs. HerbaceousVegetation: both display dense green textures; Forest images tend to be more homogeneous.
- Highway vs. River: both classes produce thin elongated linear structures that are easily confused at low resolution.

Conversely, SeaLake (large uniform blue areas), Industrial (regular grey rectangular structures), and River (narrow elongated water signatures) stand out as visually distinctive and easy to classify correctly.

Figure 3.2 — One representative sample image per class (left) and the per-class mean image (right), computed by averaging all images belonging to each category. Mean images expose dominant large-scale patterns and highlight visually ambiguous class pairs.

3.4 Data Quality Assessment

A comprehensive data quality assessment was performed prior to any modelling. The results are summarised in Table 3.2.

Check	Result	Implication
Corrupt / failed to load	0 images	No data loss — full 27,000 images available
Wrong shape ($\neq 3 \times 64 \times 64$)	0 images	All images conform to expected tensor format
Blank (all-black or all-white)	0 images	No uninformative samples detected
Outliers (top 5% by pixel dist.)	1,350 images	Correctly labelled; variability from lighting only
Class imbalance	None detected	Accuracy is a reliable evaluation metric

Table 3.2 — Data quality assessment results for the full EuroSAT dataset (27,000 images). No corrective action was required prior to modelling.

All 27,000 images loaded successfully and conformed to the expected shape of 3×64×64. Outlier detection (pixel-wise distance from class mean) flagged 1,350 images in the top 5% of distances; manual inspection confirmed these images were correctly labelled and reflected natural variation in illumination conditions rather than annotation errors. The Highway and Residential classes exhibited the highest intra-class variability, which anticipated lower per-class performance during model evaluation.

4. Baseline CNN (Trained from Scratch)

4.1 Architecture

A compact custom CNN was designed from scratch to serve as a performance baseline. The architecture consists of three convolutional blocks, followed by global average pooling, dropout, and a linear classification head. The design was deliberately kept simple — the goal was to establish a reasonable reference point rather than maximise performance.

Key design principles:

- Filter doubling ($32 \rightarrow 64 \rightarrow 128$): enables the network to learn increasingly abstract representations with depth.
- Batch Normalisation after each convolution: stabilises gradient flow and improves convergence speed.
- Max Pooling (2×2): progressively halves the spatial dimensions after each block.
- Global Average Pooling: replaces fully connected layers to reduce the number of parameters and risk of overfitting.
- Dropout ($p=0.3$): applied before the final classifier to regularise training.

Layer	Type	Output Shape	Parameters
conv1	Conv2d ($3 \rightarrow 32$, 3×3)	(1, 32, 64, 64)	896
bn1	BatchNorm2d	(1, 32, 64, 64)	64
pool	MaxPool2d (2×2)	(1, 32, 32, 32)	—
conv2	Conv2d ($32 \rightarrow 64$, 3×3)	(1, 64, 32, 32)	18,496
bn2	BatchNorm2d	(1, 64, 32, 32)	128
pool	MaxPool2d (2×2)	(1, 64, 16, 16)	—
conv3	Conv2d ($64 \rightarrow 128$, 3×3)	(1, 128, 16, 16)	73,856
bn3	BatchNorm2d	(1, 128, 16, 16)	256
pool	MaxPool2d (2×2)	(1, 128, 8, 8)	—
gap	AdaptiveAvgPool2d	(1, 128)	—
dropout	Dropout ($p=0.3$)	(1, 128)	—
classifier	Linear ($128 \rightarrow 10$)	(1, 10)	1,290
TOTAL	Trainable Parameters	—	94,986

Table 4.1 — Layer-by-layer summary of the baseline CNN architecture. Total trainable parameters: 94,986.

Figure 4.1 — Forward pass diagram of the baseline CNN. Input: $3 \times 64 \times 64$. Output: 10-class softmax probabilities.

4.2 Hyperparameter Search

A systematic hyperparameter search was conducted to identify the optimal training configuration. The full search space is presented in Table 4.2.

Hyperparameter	Values Explored
Learning rate	1×10^{-2} , 1×10^{-3} , 1×10^{-4}
Batch size	32, 64, 128
Optimiser	SGD, Adam, RMSprop
Number of conv. layers	3, 4, 5
Dropout rate	With / without; 0.2, 0.3, 0.5
Batch normalisation	With / without
Data augmentation	Type and intensity (flips, rotations)

Table 4.2 — Hyperparameter space explored during the baseline CNN search.

Three candidate configurations were trained and evaluated. Results are presented in Table 4.3.

Run	Configuration	Epochs	Val Acc	Outcome
1	Adam / lr=0.001	12	86.85%	Best — smooth convergence
2	Adam / lr=0.01	4	75.53%	Oscillates — LR too high
3	RMSprop / lr=0.005	7	54.05%	Early stop — overfit

Table 4.3 — Hyperparameter search results for the baseline CNN. Run 1 was selected as the best configuration based on validation accuracy and convergence stability.

Run 1 (Adam, lr=0.001) converged smoothly over 12 epochs with limited oscillation, achieving the highest validation accuracy of 86.8563%. Run 2 (Adam, lr=0.01) was excessively aggressive, inducing validation loss oscillations. Run 3 (RMSprop, lr=0.005) triggered early stopping after only three epochs, indicating poor generalisation capacity.

Figure 4.2 — Training and validation loss and accuracy curves for the three hyperparameter configurations. Run 1 demonstrates stable, monotonic convergence.

4.3 Results

The best configuration (Run 1: Adam, lr=0.001, 12 epochs) achieved a test accuracy of 86.85% on the held-out validation split. The best-performing classes were Industrial and Residential, while Highway was the hardest, frequently confused with River — a result consistent with the inter-class ambiguity identified during EDA.

Figure 4.3 — Confusion matrix for the baseline CNN on the validation set. Note the systematic confusion between Highway and River, and between AnnualCrop and PermanentCrop, consistent with EDA predictions.

While the 86.85% accuracy represents a solid starting point for a 94K-parameter model trained entirely from scratch, it confirmed the hypothesis that learning discriminative features for satellite imagery from limited data is fundamentally challenging. Transfer learning was therefore expected to yield material improvements.

5. Transfer Learning

Transfer learning exploits a fundamental property of deep convolutional networks: visual features learned on large general-purpose datasets — such as edges, textures, and shapes — are sufficiently universal to be repurposed for more specialised tasks. Rather than training from random initialisation, a model pre-trained on millions of images is adapted to the target domain by replacing and training only the final classification head.

Two pretrained models were selected: Microsoft ResNet-50 (He et al., 2015) and Facebook/Meta ConvNeXT-tiny (Liu et al., 2022). Both are CNN architectures pre-trained on ImageNet, but differ substantially in their architectural philosophy and era of development.

5.1 ResNet-50

5.1.1 Architecture and Methodology

ResNet-50, introduced by He et al. in 2015, is a 50-layer deep residual network comprising 23.5 million parameters. Its defining characteristic is the use of skip connections, which allow gradients to propagate directly across multiple layers, effectively resolving the vanishing gradient problem that limits the practical depth of naive deep networks. The network is organised into four residual stages with progressively wider feature maps (64 → 128 → 256 → 512 channels).

To adapt ResNet-50 to EuroSAT, three steps were followed:

- Head replacement: the original 1,000-class ImageNet classifier was replaced with a new 10-class fully connected layer matching the EuroSAT classes.
- Backbone freezing: all convolutional weights were frozen; only the new classification head was trained.
- Preprocessing: images were resized from 64×64 to 224×224 pixels and normalised using the ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

5.1.2 Results

ResNet-50 was trained for 10 epochs using the Adam optimiser (lr=0.001). The model achieved a training accuracy of 99.00% and a test accuracy of 96.26%. Training loss decreased monotonically from 0.3318 at epoch 1 to 0.0332 at epoch 9, confirming stable convergence.

Class	Precision	Recall	F1-Score
AnnualCrop	96%	98%	97%
Forest	99%	98%	98%
HerbaceousVegetation	91%	97%	94%
Highway	97%	99%	98%
Industrial	91%	100%	95%
Pasture	94%	97%	96%
PermanentCrop	97%	84%	90%
Residential	98%	94%	96%
River	100%	95%	97%
SeaLake	99%	100%	99%

Class	Precision	Recall	F1-Score
Weighted Avg	96%	96%	96%

Table 5.1 — Per-class classification report for ResNet-50 on the EuroSAT test set. Weighted average F1-score: 96%.

Performance was highly consistent across all classes, with F1-scores ranging from 90% (PermanentCrop) to 99% (SeaLake). SeaLake achieves near-perfect classification owing to its highly distinctive spectral and spatial signature in satellite imagery. PermanentCrop shows the weakest performance, with 74 images misclassified as HerbaceousVegetation, which is consistent with the strong visual similarity between these two rural land-cover types.

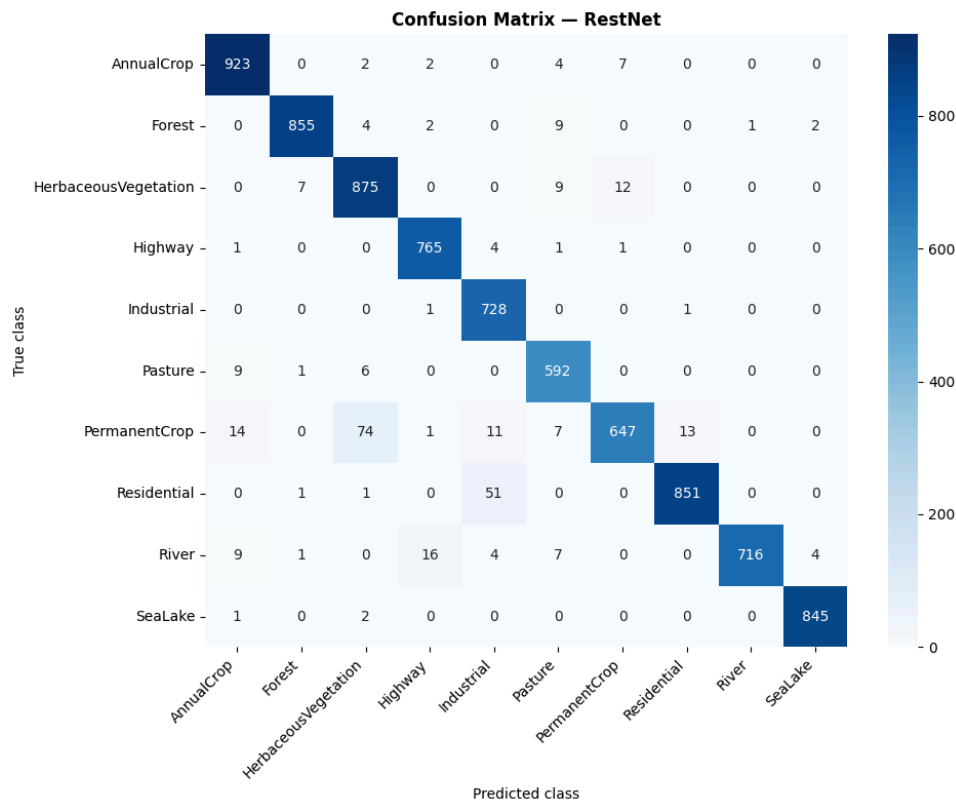


Figure 5.1 — Confusion matrix for ResNet-50 on the EuroSAT validation set. Dominant diagonal indicates strong classification across all classes, with PermanentCrop/HerbaceousVegetation as the primary confusion pair.

5.2 ConvNeXT-tiny

5.2.1 Architecture and Methodology

ConvNeXT-tiny was introduced by Liu et al. at Facebook AI Research in 2022. It represents a modernisation of the classical CNN paradigm, incorporating design insights from Vision Transformers to improve expressiveness without abandoning the convolutional inductive bias. Key architectural innovations include larger 7×7 depthwise kernels (versus 3×3 in ResNet), layer normalisation in place of batch normalisation, GELU activations instead of ReLU, and an inverted bottleneck structure with wider intermediate layers.

The same three-step adaptation procedure applied to ResNet-50 was used for ConvNeXT-tiny: head replacement, backbone freezing, and preprocessing with model-specific ImageNet statistics. Crucially, only 7,960 parameters (the classification head) were trained.

5.2.2 Results

ConvNeXT-tiny was trained for 5 epochs using the Adam optimiser ($\text{lr}=0.001$). The model achieved a training accuracy of 96.25% and a test accuracy of 95.16% — within 1.1 percentage points of ResNet-50 — with half the number of training epochs and roughly 3,000 times fewer trainable parameters.

Class	Precision	Recall	F1-Score
AnnualCrop	96%	95%	96%
Forest	98%	97%	97%
HerbaceousVegetation	93%	96%	94%
Highway	95%	90%	92%
Industrial	95%	98%	97%
Pasture	90%	94%	92%
PermanentCrop	94%	91%	93%
Residential	98%	96%	97%
River	93%	93%	93%
SeaLake	98%	99%	99%
Weighted Avg	95%	95%	95%

Table 5.2 — Per-class classification report for ConvNeXT-tiny on the EuroSAT test set. Weighted average F1-score: 95%.

The primary confusion pairs were Highway/River (39 misclassifications), consistent with the visual similarity of thin linear structures at satellite resolution, and Pasture, which received 61 incorrect classifications across several classes. The 10 most confident incorrect predictions were visually inspected and confirmed to represent genuinely ambiguous images that would challenge human annotators as well.

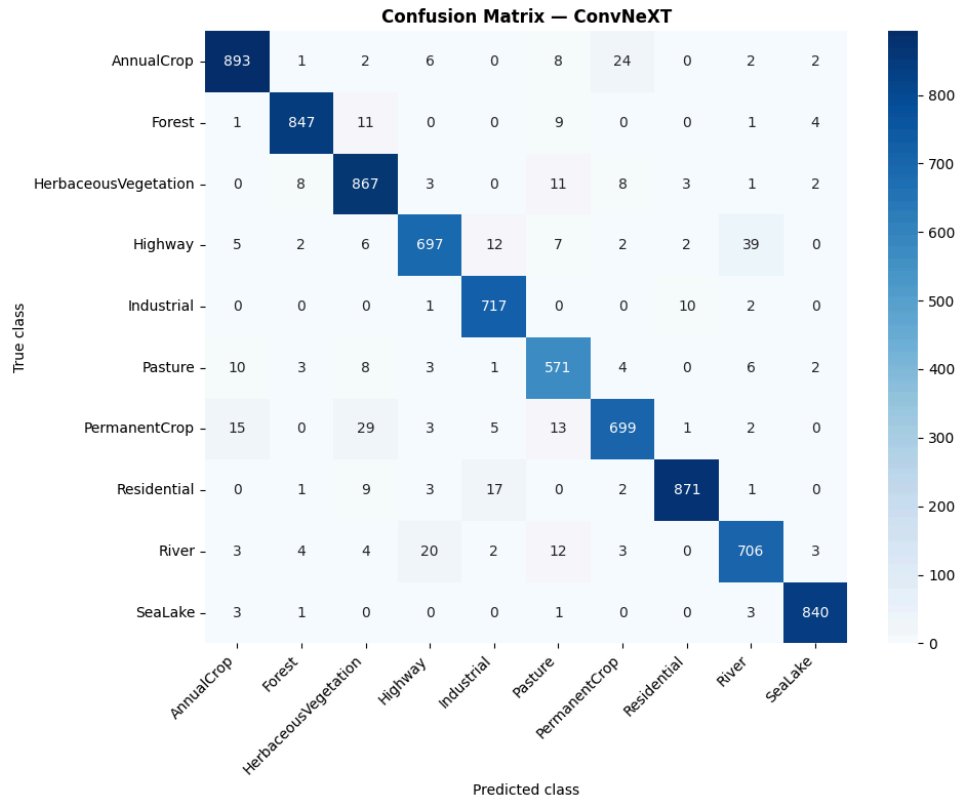


Figure 5.2 — Confusion matrix for ConvNeXT-tiny on the EuroSAT validation set.



Figure 5.3 — Ten most confident incorrect predictions from ConvNeXT-tiny. Most errors are visually understandable: PermanentCrop predicted as AnnualCrop, Highway predicted as River.

6. Model Comparison and Selection

All three models were evaluated across three dimensions: predictive performance, computational efficiency, and practical deployment suitability.

6.1 Performance and Efficiency Summary

Metric	Baseline CNN	ResNet-50	ConvNeXT-tiny
Test Accuracy	86.85%	96.26%	95.16%
Total Parameters	94,986	23,528,522	27,800,000
Trainable Parameters	94,986	23,528,522	7,960
Epochs Required	12	10	5
Architecture	Custom CNN	Residual (2015)	Modern CNN (2022)
Pre-training Dataset	None	ImageNet	ImageNet
Input Size	64×64 px	224×224 px	224×224 px
Relative Cost (params)	1×	248×	0.08×

Table 6.1 — Side-by-side comparison of all three models across performance, efficiency, and architecture dimensions.

Figure 6.1 — Training loss and accuracy curves for all three models. ConvNeXT-tiny achieves convergence in 5 epochs; ResNet-50 requires 10; the baseline CNN requires 12 but plateaus at a substantially lower accuracy.

6.2 Discussion

The transfer learning models demonstrate a clear advantage over the from-scratch baseline, confirming the value of leveraging large-scale pretraining for satellite image classification. The improvement of approximately 15 percentage points in test accuracy (86.85% → 96.26%) illustrates that ImageNet-pretrained features — learned from millions of diverse natural images — provide a far richer visual vocabulary than can be acquired from EuroSAT's 18,900 training images alone.

Between the two pretrained models, the difference in test accuracy is minimal: 96.26% (ResNet-50) versus 95.16% (ConvNeXT-tiny). However, ResNet-50 requires 23,528,522 trainable parameters and 10 epochs to achieve this result, whereas ConvNeXT-tiny requires only 7,960 trainable parameters and 5 epochs — a reduction of approximately 3,000× in trainable parameters and 2× in training time. This efficiency advantage is decisive for resource-constrained environments and large-scale deployment.

ResNet-50 is considered a mature, well-established architecture with extensive community support and tooling. However, its 2015 design predates many advances in CNN architecture that have emerged from the Vision Transformer literature. ConvNeXT-tiny, developed in 2022, explicitly incorporates these advances into a convolutional framework, yielding a modern and highly efficient model that outperforms older baselines by design.

6.3 Final Model Recommendation

ConvNeXT-tiny is the recommended model for deployment. It achieves near-identical test accuracy to ResNet-50 (95.16% vs. 96.26%) while requiring approximately 3,000× fewer trainable parameters and half the training epochs. Its modern architecture, computational efficiency, and strong generalisation performance make it the optimal choice for production use cases, particularly in resource-constrained or latency-sensitive environments.

ResNet-50 remains a valid alternative in scenarios where the marginal 1.1% accuracy improvement is operationally significant and computational cost is not a constraint.

7. Conclusion

This study evaluated three deep learning approaches for land-cover classification on the EuroSAT benchmark. The key findings are as follows:

- Dataset: EuroSAT is well-balanced, clean, and requires no corrective preprocessing. All 27,000 images loaded correctly with expected shape and no missing or blank files.
 - Baseline CNN: the custom 3-block CNN achieved 86.85% test accuracy using 94,986 parameters trained from scratch over 12 epochs. Strongest classes: Industrial and Residential (F1 = 0.79). Hardest class: Highway (F1 = 0.33).
 - ResNet-50: achieved 96.26% test accuracy — a gain of approximately 9.4 percentage points over the baseline — using 23.5 million trainable parameters over 10 epochs.
 - ConvNeXT-tiny: achieved 95.16% test accuracy with only 7,960 trainable parameters in 5 epochs, demonstrating that a modern frozen backbone can rival a full ResNet at a fraction of the computational cost.
 - Recommendation: ConvNeXT-tiny is the preferred production model, offering near-identical accuracy to ResNet-50 with substantially lower resource requirements.
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8. Future Work

Several directions for future investigation are identified:

- Progressive fine-tuning: gradually unfreezing the last 2–3 convolutional blocks of ConvNeXT-tiny and ResNet-50 with a reduced learning rate (e.g. 1×10^{-5}) may push test accuracy above 97% without a substantial increase in computational cost.
- Targeted augmentation: Highway (F1 = 0.92) and PermanentCrop (F1 = 0.93) remain the weakest classes. Class-specific augmentation strategies — such as rotation, colour jitter, and random cropping — could improve generalisation on these visually ambiguous categories.
- Expanded hyperparameter search: the current search covered three configurations. A broader grid or random search over batch size, weight decay, and dropout rate would increase confidence in the optimal configuration.
- Vision Transformer evaluation: evaluating Vision Transformer (ViT) and hybrid architectures would allow direct comparison against the most recent state-of-the-art models for remote sensing classification.

- Out-of-distribution testing: EuroSAT covers a specific set of European geographical regions. Testing on satellite imagery from other regions or seasons would quantify the generalisation of learned features beyond the training distribution.
- Multispectral extension: EuroSAT also provides multispectral (13-band) imagery. Incorporating all spectral channels may yield further accuracy improvements for certain land-cover categories.
- Edge deployment: applying model quantisation and pruning to the ConvNeXT-tiny model would assess suitability for real-time inference on resource-constrained edge hardware.
- Cloud deployment: deploying the best model as a REST API or serverless function for scalable, on-demand satellite image classification.

References

- [1] Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217–2226.
- [2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- [3] Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11976–11986.
- [4] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). ImageNet: A Large-Scale Hierarchical Image Database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 248–255.
- [5] Paszke, A., Gross, S., Massa, F., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 32.
- [6] Wolf, T., Debut, L., Sanh, V., et al. (2020). Transformers: State-of-the-Art Natural Language Processing. In *Proceedings of EMNLP: System Demonstrations*, 38–45. (Hugging Face Transformers library.)